05 Continuity

1. CONTINUOUS FUNCTIONS:

A function for which a small change in the independent variable causes only a small change and not a sudden jump in the dependent variable are called continuous functions. Naively, we may say that a function is continuous at a fixed point if we can draw the graph of the function around that point without lifting the pen from the plane of the paper.

A function $f(x)$ is said to be continuous at $x = a$, if $\lim\limits_{x\to a} f(x)$ exists and is equal to $f(a)$. Symbolically

$f(x)$ is continuous at $x = a$ if $\lim\limits_{h\to 0} f(a-h) = \lim\limits_{h\to 0} f(a+h) = f(a) =$ finite quantity.

i.e. $\text{LHL}|_{x=a}$ = $\text{RHL}|_{x=a}$ = value of $f(x)|_{x=a}$ = finite quantity. $(h > 0)$

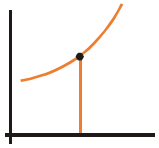


figure (1)

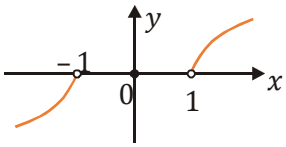


figure (2)

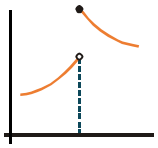


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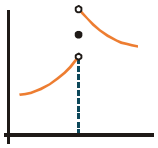


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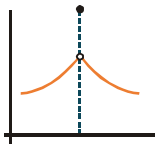


figure (5)

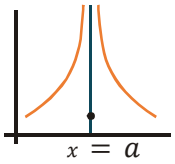


figure (6)

In figure (1) and (2) $f(x)$ is continuous at $x = a$ and $x = 0$ respectively and in figure (3) to (6) $f(x)$ is discontinuous at $x = a$.

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Note 1 : Continuity of a function must be discussed only at points which are in the domain of the function.

Note 2 : If $x = a$ is an isolated point of domain then $f(x)$ is always considered to be continuous at $x = a$.

Illustration 1

If $f(x) = \begin{cases} \sin\frac{\pi x}{2}, & x<1 \\ [x] & x\geq 1 \end{cases}$ then find whether $f(x)$ is continuous or not at $x = 1$, where [.] denotes greatest integer function.

Solution:

$$f(x) = \begin{cases} \sin\frac{\pi x}{2}, & x<1 \\ [x], & x\geq 1 \end{cases}$$

For continuity at $x = 1$, we determine, $f(1)$ $\lim\limits_{x\to 1^-} f(x)$ and $\lim\limits_{x\to 1^+} f(x)$.

Now, $f(1) = [1] = 1$

$\lim\limits_{x\to 1^-} f(x) = \lim\limits_{x\to 1^-} \sin\frac{\pi x}{2} = \sin\frac{\pi}{2} = 1$ and $\lim\limits_{x\to 1^+} f(x) = \lim\limits_{x\to 1^+} [x] = 1$

so $f(1) = \lim\limits_{x\to 1^-} f(x) = \lim\limits_{x\to 1^+} f(x)$

$\therefore$ $f(x)$ is continuous at $x = 1$

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Illustration 2

Let $f(x)=\begin{cases}\dfrac{a(1-x\sin x)+b\cos x+5}{x^2} & x<0\\ 3 & x=0\\ \left(1+\left(\dfrac{cx+dx^3}{x^2}\right)\right)^{\frac{1}{x}} & x>0\end{cases}$

If f is continuous at $x=0$, then find out the values of a, b, c and d.

Solution:

Since $f(x)$ is continuous at $x=0$, so at $x=0$, both left and right limits must exist and both must be equal to 3.

Now $\lim\limits_{x\to0^-}\dfrac{a(1-x\sin x)+b\cos x+5}{x^2}=\lim\limits_{x\to0^-}\dfrac{(a+b+5)+\left(-a-\dfrac{b}{2}\right)x^2+\ldots}{x^2}=3$ (By the expansions of sin$x$ and cos$x$)

If $\lim\limits_{x\to0^-}f(x)$ exists then $a+b+5=0$ and $-a-\dfrac{b}{2}=3\Rightarrow a=-1$ and $b=-4$

since $\lim\limits_{x\to0^+}\left(1+\left(\dfrac{cx+dx^3}{x^2}\right)\right)^{\frac{1}{x}}$ exists $\Rightarrow\lim\limits_{x\to0^+}\dfrac{cx+dx^3}{x^2}=0\Rightarrow c=0$

Now $\lim\limits_{x\to0^+}(1+dx)^{\frac{1}{x}}=\lim\limits_{x\to0^+}\left[(1+dx)^{\frac{1}{dx}}\right]^d=e^d$

So $e^d=3\Rightarrow d=\ell n\,3$,

Hence $a=-1, b=-4, c=0$ and $d=\ell n\,3$.

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2. CONTINUITY OF THE FUNCTION IN AN INTERVAL:

(a) A function is said to be continuous in (a,b) if f is continuous at each & every point belonging to (a,b).

(b) A function is said to be continuous in a closed interval $[a,b]$ if :

(i) f is continuous in the open interval (a,b)

(ii) f is right continuous at 'a' i.e. $\lim\limits_{x\to a^+}f(x)=f(a)$ = a finite quantity

(iii) f is left continuous at 'b' i.e. $\lim\limits_{x\to b^-}f(x)=f(b)$ = a finite quantity

Note:

(i) All polynomials, trigonometrical functions, exponential & logarithmic functions are continuous in their domains.

(ii) If $f(x)$ & $g(x)$ are two functions that are continuous at $x=c$ then the function defined by : $F_1(x)=f(x)\pm g(x)$; $F_2(x)=K\ f(x)$,where K is any real number $F_3(x)=f(x).g(x)$; are also continuous at $x=c$.

Further, if $g(c)$ is not zero, then $F_4(x)=\dfrac{f(x)}{g(x)}$ is also continuous at $x=c$.

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Illustration 3

If $f(x)=\begin{cases}\dfrac{\sin(p+1)+\sin x}{x} & , \quad x<0\\ q & , \quad x=0\\ \dfrac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}} & , \quad x>0\end{cases}$ is continuous at $x = 0$, then the ordered pair (p, q) is equal to:

(A) $\left(\frac{5}{2},\frac{1}{2}\right)$ (B) $\left(-\frac{3}{2},-\frac{1}{2}\right)$ (C) $\left(-\frac{1}{2},\frac{3}{2}\right)$ (D) $\left(-\frac{3}{2},\frac{1}{2}\right)$

Ans. (D)

Solution:

$$\text{RHL}=\lim_{x\to0^+}\frac{\sqrt{x+x^2}-\sqrt{x}}{x^{\frac{3}{2}}}=\lim_{x\to0^+}\frac{\sqrt{1+x}-1}{x}=\frac{1}{2}$$

$$\text{LHL}=\lim_{x\to0}\frac{\sin(p+1)x+\sin x}{x}=(p+1)+1=p+2$$

for continuity LHL = RHL = $f(0)$

$\Rightarrow (p, q) = \left(\frac{-3}{2},\frac{1}{2}\right)$

Illustration 4

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If the function f defined on $\left(-\frac{1}{3},\frac{1}{3}\right)$ by $f(x)=\begin{cases}\frac{1}{x}\log_e\left(\frac{1+3x}{1-2x}\right), & \text{when } x\neq0\\ k & \text{, when } x=0\end{cases}$ is continuous, then k is equal to_____

Ans. (5.00)

Solution:

$$k=\lim_{x\to0}\left(\frac{\ell n(1+3x)}{x}-\frac{\ell n(1-2x)}{x}\right)$$

$k = 3 + 2 = 5$

Illustration 5

If $f(x)=\begin{cases}\dfrac{\sin(a+2)x+\sin x}{x} & ; \quad x<0\\ b & ; \quad x=0\\ \dfrac{(x+3x^2)^{\frac{1}{3}}-x^{-\frac{1}{3}}}{x^{\frac{4}{3}}} & ; \quad x>0\end{cases}$ is continuous at $x = 0$, then $a + 2b$ is equal to :

(A) –1 (B) 1 (C) –2 (D) 0

Ans. (D)

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Solution:

$$\lim_{x\to 0^-} f(x)=\lim_{x\to 0}\left(\frac{\sin(a+2)x}{x}+\frac{\sin x}{x}\right)=a+3$$

$$\lim_{x\to 0^-} f(x)=\lim_{x\to 0}\frac{(x+3x^2)^{1/3}-x^{1/3}}{x^{4/3}}$$

$$=\lim_{x\to 0}\frac{(1+3x)^{1/3}-1}{x}=1$$

$f(0) = b$

for continuity at $x = 0$

$$\lim_{x\to 0^-} f(x)= f(0)=\lim_{x\to 0^+} f(x)$$

$\Rightarrow a + 3 = b = 1$

$\therefore a = -2, b = 1$

$\therefore a + 2b = 0$

Illustration 6

Let $f : R \to R$ be a function defined as $f(x)=\begin{cases}\dfrac{\sin(a+1)x+\sin 2x}{2x} & , \text{if } x<0\\ b & , \text{if } x=0\\ \dfrac{\sqrt{x+bx^3}-\sqrt{x}}{bx^{5/2}} & , \text{if } x>0\end{cases}$. If f is continuous at $x = 0$,

then the value of $a + b$ is equal to :

(A) $-\frac{5}{2}$ (B) -2 (C) -3 (D) $-\frac{3}{2}$

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Ans. (D)

Solution:

$f(x)$ is continuous at $x = 0$

$$\lim_{x\to 0^+} f(x)= f(0)=\lim_{x\to 0^-} f(x) \quad ...(1)$$

$f(0) = b$ ...(2)

$$\lim_{x\to 0^+} f(x)=\lim_{x\to 0^+}\left(\frac{\sin(a+1)x}{2x}+\frac{\sin 2x}{2x}\right)$$

$$=\frac{a+1}{2}+1 \quad ...(3)$$

$$\lim_{x\to 0^+} f(x)=\lim_{x\to 0^+}\frac{\sqrt{x+bx^3}-\sqrt{x}}{bx^{5/2}}$$

$$=\lim_{x\to 0^+}\frac{(x+bx^3-x)}{bx^{5/2}\left(\sqrt{x+bx^3}+\sqrt{x}\right)}$$

$$=\lim_{x\to 0^+}\frac{\sqrt{x}}{\sqrt{x}\left(\sqrt{1+bx^2}+1\right)}=\frac{1}{2} \quad ...(4)$$

Use (2), (3) & (4) in (1)

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$\frac{1}{2}=b=\frac{a+1}{2}+1$

$\Rightarrow b=\frac{1}{2}, a=-2$

$a+b=\frac{-3}{2}$

Illustration 7

Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=\begin{cases}\frac{x^3}{(1-\cos 2x)^2}\log_e\left(\frac{1+2xe^{-2x}}{\left(1-xe^{-x}\right)^2}\right) & , \quad x\neq 0\\ \alpha & , \quad x=0\end{cases}$. If f is continuous at $x=0$, then α is equal to :

(A) 1 (B) 3 (C) 0 (D) 2

Ans. (A)

Solution:

For continuity

$\lim\limits_{x\to 0}\frac{x^3}{4\sin^4 x}\left(\ell n(1+2xe^{-2x})-2\ell n(1-xe^{-x})\right)$

$=\alpha$

$\lim\limits_{x\to 0}\frac{1}{4x}[2xe^{-2x}+2xe^{-x}]=\alpha$

$=\frac{1}{4}(4)=\alpha=1$

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Illustration 8

Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=\begin{cases}\frac{\lambda|x^2-5x+6|}{\mu(5x-x^2-6)}, & x<2\\ e^{\frac{\tan(x-2)}{x-[x]}} & ,x>2\\ \mu & ,x=2\end{cases}$, where $[x]$ is the greatest integer less than or equal to x. If f is continuous at $x=2$, then $\lambda+\mu$ is equal to :

(A) $e(-e+1)$ (B) $e(e-2)$ (C) 1 (D) $2e-1$

Ans. (A)

Solution:

$\lim\limits_{x\to 2^+} f(x)=\lim\limits_{x\to 2^+} e^{\frac{\tan(x-2)}{x-2}}=e^1$

$\lim\limits_{x\to 2^-} f(x)=\lim\limits_{x\to 2^-}\frac{-\lambda(x-2)(x-3)}{\mu(x-2)(x-3)}=-\frac{\lambda}{\mu}$

For continuity $\mu=e=-\frac{\lambda}{\mu}\Rightarrow\mu=e,\lambda=-e^2$

$\lambda+\mu=e(-e+1)$

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Illustration 9

Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=\begin{cases}2\sin\left(-\frac{\pi x}{2}\right), & \text{if } x<-1\\ |ax^2+x+b|, & \text{if } -1\le x\le 1\\ \sin(\pi x), & \text{if } x>1\end{cases}$. If $f(x)$ is continuous on R, then $a+b$ equals:

(A) –3 (B) –1 (C) 3 (D) 1

Ans. (B)

Solution:

$f(x)$ is continuous on R

$\Rightarrow f(1^-)=f(1)=f(1^+)$

$|a+1+b|=\lim\limits_{x\to1}\sin(\pi x)$

$|a+1+b|=0\Rightarrow a+b=-1$ ...(1)

$\Rightarrow$ Also $f(-1^-)=f(-1)=f(-1^+)$

$\lim\limits_{x\to-1}2\sin\left(\frac{-\pi x}{2}\right)=|a-1+b|$

$|a-1+b|=2$

Either $a-1+b=2$ or $a-1+b=-2$

$a+b=3$ ...(2)

or $a+b=-1$ ...(3)

from (1) and (2) $\Rightarrow a+b=3=-1$(reject)

from (1) and (3) $\Rightarrow a+b=-1$

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Illustration 10

Let $a, b\in\mathbb{R}, b\neq0$, Define a function $f(x)=\begin{cases}a\sin\frac{\pi}{2}(x-1), & \text{for } x\le0\\ \frac{\tan2x-\sin2x}{bx^3}, & \text{for } x>0.\end{cases}$. If f is continuous at $x=0$, then $10-ab$ is equal to _____.

Ans. (14)

Solution:

$$f(x)=\begin{cases}a\sin\frac{\pi}{2}(x-1), & x\le0\\ \frac{\tan2x-\sin2x}{bx^3}, & x>0\end{cases}$$

For continuity at '0'

$\lim\limits_{x\to0^+}f(x)=f(0)$

$\Rightarrow \lim\limits_{x\to0^+}\frac{\tan2x-\sin2x}{bx^3}=-a$

$\Rightarrow \lim\limits_{x\to0^+}\frac{\frac{8x^3}{3}+\frac{8x^3}{3!}}{bx^3}=-a$

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$\Rightarrow 8\left(\frac{1}{3}+\frac{1}{3!}\right)=-ab$

$\Rightarrow 4=-ab$

$\Rightarrow 10-ab=14$

Illustration 11

Let $f:\left(-\frac{\pi}{4},\frac{\pi}{4}\right)\to\mathbb{R}$ be defined as $f(x)=\begin{cases}(1+|\sin x|)^{\frac{3a}{|\sin x|}} & , \quad -\frac{\pi}{4}<x<0\\ b & , \quad x=0\\ e^{\cot 4x/\cot 2x} & , \quad 0<x<\frac{\pi}{4}\end{cases}$.

If f is continuous at $x=0$, then the value of $6a+b^2$ is equal to:

(A) $1-e$ (B) $e-1$ (C) $1+e$ (D) e

Ans. (C)

Solution:

$\lim\limits_{x\to 0} f(x)=b$

$\lim\limits_{x\to 0^+} xe^{\frac{\cot 4x}{\cot 2x}}=e^{\frac{1}{2}}=b$

$\lim\limits_{x\to 0^-}(1+|\sin x|)^{\frac{3a}{|\sin x|}}=e^{3a}=e^{\frac{1}{2}}$

$\lim\limits_{x\to 0^-}(1+|\sin x|)^{\frac{3a}{|\sin x|}}=e^{3a}=e^{\frac{1}{2}}$

$a=\frac{1}{6}\Rightarrow 6a=1$

$(6a+b^2)=(1+e)$

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3. THE INTERMEDIATE VALUE THEOREM:

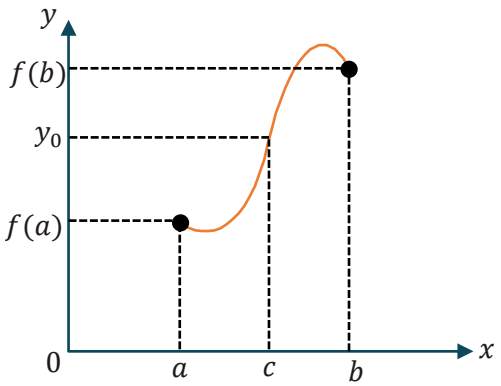
Suppose $f(x)$ is continuous on an interval I, and a and b are any two points of I. Then if y_0 is a number between $f(a)$ and $f(b)$, there exists a number c between a and b such that $f(c)=y_0$

Note that a function f which is continuous in $[a,b]$ possesses the following properties :

(i) If $f(a)$ & $f(b)$ posses opposite signs, then there exists atleast one root of the equation $f(x)=0$ in the open interval (a,b).

(ii) If K is any real number between $f(a)$ & $f(b)$, then there exists atleast one root of the equation $f(x)=K$ in the open interval (a,b).

Note: In above cases the number of roots is always odd.



The function f, being continuous on $[a,b]$ takes on every value between $f(a)$ and $f(b)$

Illustration 12

Show that the function, $f(x)=(x-a)^2(x-b)^2+x$, takes the value $\frac{a+b}{2}$ for some $x_0\in(a,b)$

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Solution:

$f(x) = (x-a)^2(x-b)^2 + x$

$f(a) = a$

$f(b) = b$

& $\frac{a+b}{2} \in (f(a), f(b))$

$\therefore$ By intermediate value theorem, there is atleast one $x_0 \in (a,b)$ such that $f(x_0) = \frac{a+b}{2}$.

Illustration 13

Let $f: [0,1] \xrightarrow{onto} [0,1]$ be a continuous function, then prove that $f(x) = x$ for atleast one $x \in [0,1]$

Solution:

Consider $g(x) = f(x) - x$

$g(0) = f(0) - 0 = f(0) \geq 0$ $\quad \{\because \quad 0 \leq f(x) \leq 1\}$

$g(1) = f(1) - 1 \leq 0$

$\Rightarrow g(0).g(1) \leq 0$

$\Rightarrow g(x) = 0$ has atleast one root in $[0,1]$

$\Rightarrow f(x) = x$ for atleast one $x \in [0,1]$

4. SOME IMPORTANT POINTS:

(a) If $f(x)$ is continuous & $g(x)$ is discontinuous at $x = a$ then the product function $\phi(x) = f(x).g(x)$ will **not necessarily be discontinuous at $x = a$**, e.g.

$$f(x) = x \,\&\, g(x) = \begin{cases} \sin\frac{\pi}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$f(x)$ is continuous at $x = 0$ & $g(x)$ is discontinuous at $x = 0$, but $f(x).g(x)$ is continuous at $x = 0$.

(b) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then the product function $\phi(x) = f(x).g(x)$ **is not necessarily be discontinuous at $x = a$**, e.g.

$$f(x) = -g(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

$f(x)$ & $g(x)$ both are discontinuous at $x = 0$ but the product function $f.g(x)$ is still continuous at $x = 0$

(c) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then $f(x) \pm g(x)$ is not necessarily be discontinuous at $x = a$

(d) A continuous function whose domain is closed must have a range also in closed interval.

(e) If f is continuous at $x = a$ & g is continuous at $x = f(a)$ then the composite $g[f(x)]$ is continuous at $x = a$. eg. $f(x) = \frac{x\sin x}{x^2+2}$ & $g(x) = |x|$ are continuous at $x = 0$, hence the composite

$(gof)(x) = \left|\frac{x\sin x}{x^2+2}\right|$ will also be continuous at $x = 0$

Illustration 14

If $f(x) = \frac{x+1}{x-1}$ and $g(x) = \frac{1}{x-2}$, then discuss the continuity of $f(x)$, $g(x)$ and $fog(x)$ in its domain.

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Solution:

$f(x)=\dfrac{x+1}{x-1}$

$f(x)$ is a rational function it must be continuous in its domain and f is not defined at $x = 1$.

$g(x)=\dfrac{1}{x-2}$

$g(x)$ is also a rational function. It must be continuous in its domain and g is not defined at $x = 2$.

Consider $g(x) = 1$

$\dfrac{1}{x-2} = 1 \Rightarrow x = 3$

$\therefore fog(x)$ is continuous in its domain : $\mathbb{R}$ – {2, 3}

Illustration 15

If $f(x) = \begin{cases} x+a, & x\le 0 \\ |x-4|, & x>0 \end{cases}$ and $g(x) = \begin{cases} x+1, & x<0 \\ (x-4)^2+b, & x\ge 0 \end{cases}$ are continuous on R, then (gof) (2) + (fog) (–2) is equal to:

(A) –10 (B) 10 (C) 8 (D) –8

Ans. (D)

Solution:

$f(x)=\begin{cases} x+a & ; x\le 0 \\ |x-4| & ; x>0 \end{cases}$; $g(x)=\begin{cases} x+1 & ; x<0 \\ (x-4)^2+b & ; x\ge 0 \end{cases}$

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For continuity $a = 4$ and $b = -15$

$g(f(2))+f(g(-2))$

$=g(2)+f(-1)=-8$

5. CONTINUITY OVER COUNTABLE SET:

There are functions which are continuous over a countable set and else where discontinuous.

Illustration 16

If $f(x)=\begin{cases} x & \text{if } x\in \mathbb{Q} \\ -x & \text{if } x\notin \mathbb{Q} \end{cases}$, find the points where $f(x)$ is continuous

Solution:

Let $x = a$ be the point at which $f(x)$ is continuous.

$\Rightarrow \lim\limits_{\substack{x\to a \\ \text{through rational}}} f(x) = \lim\limits_{\substack{x\to a \\ \text{through irrational}}} f(x)$

$\Rightarrow a = -a$

$\Rightarrow a = 0 \Rightarrow$ function is continuous at $x = 0$.

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Illustration 17

If the function $f(x)=\begin{cases}\frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right) & , \quad x<0\\ k & , \quad x=0\\ \frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1} & , \quad x>0\end{cases}$ is continuous at $x = 0$, then $\frac{1}{a}+\frac{1}{b}+\frac{4}{k}$ is equal to:

(A) –5 (B) 5 (C) – 4 (D) 4

Ans. (A)

Solution:

If $f(x)$ is continuous at $x = 0$, RHL = LHL = $f(0)$

$\lim\limits_{x\to0^+} f(x)=\lim\limits_{x\to0^+}\frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1}.\frac{\sqrt{x^2+1}+1}{\sqrt{x^2+1}+1}$ (Rationalisation)

$\lim\limits_{x\to0^+}-\frac{2\sin^2 x}{x^2}.\left(\sqrt{x^2+1}+1\right)=-4$

$\lim\limits_{x\to0^-} f(x)=\lim\limits_{x\to0^-}\frac{1}{x}\ell n\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right)$

$\lim\limits_{x\to0^-}\frac{\ell n\left(1+\frac{x}{a}\right)}{\left(\frac{x}{a}\right).a}+\frac{\ell n\left(1-\frac{x}{b}\right)}{\left(-\frac{x}{b}\right).b}$

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$=\frac{1}{a}+\frac{1}{b}$

So $\frac{1}{a}+\frac{1}{b}=-4=k$

$\Rightarrow\frac{1}{a}+\frac{1}{b}+\frac{4}{k}=-4-1=-5$

Illustration 18

If for $p\neq q\neq0$, then function $f(x)=\frac{\sqrt[7]{p(729+x)}-3}{\sqrt[3]{729+qx}-9}$ is continuous at $x = 0$, then:

(A) $7pq\,f(0) - 1 = 0$ (B) $63q\,f(0) - p^2 = 0$

(C) $21q\,f(0) - p^2 = 0$ (D) $7pq\,f(0) - 9 = 0$

Ans. (B)

Solution:

$f(0)=\lim\limits_{x\to0} f(x)$

Limit should be $\frac{0}{0}$ form

So, $\sqrt[7]{p.729}-3=0\Rightarrow p.3^6 = 3^7\Rightarrow p = 3$

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Now, $f(0)=\lim_{x\to 0}\frac{\sqrt[7]{3(3^6+x)}-3}{\sqrt[3]{3^6+qx}-9}$

$=\lim_{x\to 0}\frac{3\left[\left(1+\frac{x}{3^6}\right)^{1/7}-1\right]}{9\left[\left(1+\frac{qx}{3^6}\right)^{1/3}-1\right]}=\frac{3}{9}\times\frac{\frac{1}{7.3^6}}{\frac{q}{3.3^6}}$

$\Rightarrow f(0)=\frac{1}{3}\times\frac{3}{7q}=\frac{1}{7q}$

$\Rightarrow 7qf(0)-1=0$

$\Rightarrow 7.p^2.qf(0)-p^2=0$ (for option)

$\Rightarrow 63qf(0)-p^2=0$

Illustration 19

Let $f:[-1,3]\to R$ be defined as $f(x)=\begin{cases}|x|+[x] & , \ -1\le x<1\\ x+|x| & , \ 1\le x<2\\ x+[x] & , \ 2\le x\le 3\end{cases}$, where $[t]$ denotes the greatest integer less than or equal to t. Then, f is discontinuous at:

(A) four or more points (B) only one point

(C) only two points (D) only three points

Ans. (D)

Solution:

$$f(x)=\begin{cases}-(x+1) & , \ -1\le x<0\\ x & , \ 0\le x<1\\ 2x & , \ 1\le x<2\\ x+2 & , \ 2\le x<3\\ x+3 & , \ x=3\end{cases}$$

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function discontinuous at $x = 0,1,3$

Illustration 20

Let $[t]$ denote the greatest integer $\le t$. The number of points where the function

$f(x)=[x]\left|x^2-1\right|+\sin\left(\frac{\pi}{[x]+3}\right)-[x+1], x\in(-2,2)$ is not continuous is _____.

Ans. (2)

Solution:

$f(x) = [x]\,|x^2 - 1| + \sin\frac{\pi}{[x+3]} - [x + 1]$

$$f(x)=\begin{cases}3-2x^2, & -2<x<-1\\ x^2, & -1\le x<0\\ \frac{\sqrt{3}}{2}+1 & 0\le x<1\\ x^2+1+\frac{1}{\sqrt{2}}, & 1\le x<2\end{cases}$$

discontinuous at $x = 0, 1$

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Illustration 21

Let $\alpha \in R$ be such that the function $f(x)=\begin{cases}\dfrac{\cos^{-1}(1-\{x\}^2)\sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3}, & x\neq 0\\ \alpha, & x=0\end{cases}$ is continuous at $x = 0$, where

$\{x\} = x - [x]$, $[x]$ is the greatest integer less than or equal to x. Then:

(A) $\alpha=\dfrac{\pi}{\sqrt{2}}$ (B) $\alpha = 0$ (C) no such α exists (D) $\alpha=\dfrac{\pi}{4}$

Ans (C)

Solution:

$\lim\limits_{x\to 0^+} f(x)= f(0)=\lim\limits_{x\to 0^-}(x)$

$\lim\limits_{x\to 0^+}\dfrac{\cos^{-1}(1-x^2)\cdot\sin^{-1}(1-x)}{x(1-x)(1+x)}$

$\lim\limits_{x\to 0^+}\dfrac{\cos^{-1}(1-x^2)}{x\cdot 1\cdot 1}\cdot\dfrac{\pi}{2}$

Let $1 - x^2 = \cos\theta$

$\dfrac{\pi}{2}\lim\limits_{x\to 0^+}\dfrac{\theta}{\sqrt{1-\cos\theta}}$

$\dfrac{\pi}{2}\lim\limits_{\theta\to 0^+}\dfrac{\theta}{\sqrt{2}\sin\dfrac{\theta}{2}}=\dfrac{\pi}{\sqrt{2}}$

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Now, $\lim\limits_{x\to 0^-}\dfrac{\cos^{-1}\left(1-(1+x)^2\right)\sin^{-1}(-x)}{(1+x)-(1+x)^3}$

$\lim\limits_{x\to 0^-}\dfrac{\dfrac{\pi}{2}\left(-\sin^{-1}x\right)}{(1+x)(2+x)(-x)}$

$\lim\limits_{x\to 0^-}\dfrac{\dfrac{\pi}{2}}{1\cdot 2}\cdot\dfrac{\sin^{-1}x}{x}=\dfrac{\pi}{4}$

$\Rightarrow$ RHL $\neq$ LHL

Function can't be continuous

$\Rightarrow$ No value of α exist

Illustration 22

If $f : R \to R$ is a function defined by $f(x)=[x-1]\cos\left(\dfrac{2x-1}{2}\right)\pi$, where [.] denotes the greatest integer function, then f is:

(A) discontinuous at all integral values of x except at $x = 1$

(B) continuous only at $x = 1$

(C) continuous for every real x

(D) discontinuous only at $x = 1$

Ans. (C)

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Solution:

For $x = n, n \in Z$

$$\text{LHL} = \lim_{x \to n^-} f(x) = \lim_{x \to n^-} [x-1]\cos\left(\frac{2x-1}{2}\right)\pi$$

$= 0$

$$\text{RHL} = \lim_{x \to n^+} f(x) = \lim_{x \to n^+} [x-1]\cos\left(\frac{2x-1}{2}\right)\pi$$

$= 0$

$f(n) = 0$

$\Rightarrow$ LHL = RHL = $f(n)$

$\Rightarrow f(x)$ is continuous for every real x

Illustration 23

The function $f(x) = [x]^2 - [x^2]$ (where [y] is the greatest integer less than or equal to y), is discontinuous at :

(A) all integers
(B) all integers except 0 and 1
(C) all integers except 0
(D) all integers except 1

Ans. (C)

Solution:

$$f(x) = [x]^2 - [x^2]$$

At $x = I$ (Integer)

$$f(I) = [I]^2 - [I^2] = I^2 - I^2 = 0$$

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$$f(I^+) = \lim_{h \to 0} [I+h]^2 - [(I+h)^2]$$

$$= I^2 - I^2 = 0$$

$$f(I^-) = \lim_{h \to 0} [I-h]^2 - [(I-h)^2]$$

$$= (I-1)^2 - (I^2 - 1) = I^2 + 1 - 2I - I^2 + 1$$

$$= 2 - 2I$$

$\Rightarrow f(x)$ is discontinuous at all integers & except 0.

$\because$ At $x = 1$,

$f(1) = f(1^+) = f(1^-) = 0$

and At $x = 0$,

$$f(0) = 0 = f(0^+)$$

$$f(0^-) = 1 =$$

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